

PROJECT FORESIGHT

Enterprise Retail Demand Forecasting & Inventory Risk Intelligence

Client: NorthBay Living

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Role: Data Science & Analytics Intern

Program: Zidio Development Internship

Repository: <https://github.com/MHK-123/Foresight>

Completion Date: August 2026

1. Executive Summary

Project FORESIGHT delivers an enterprise-grade retail demand forecasting and explainable inventory risk intelligence system for **NorthBay Living**, a retail chain operating 30 stores across 4 formats with 5,000 active SKUs. Using 4 years of operational transaction data (9.945M records), the project developed a production **LightGBM Regressor** that achieves **42.16% Out-of-Sample WAPE** (-11.3% error reduction over baseline) and **1.183 RMSE** (-20.3% error reduction) on the 2025 temporal test set.

The automated risk engine evaluated all **26,408 store-SKU positions** and identified **Rs. 1.78 Billion in trapped overstock capital** (68.1% of network) and **5,209 stockout deficit positions** with **100.0% validation recall** on ground-truth defect benchmarks. A Streamlit decision dashboard was delivered featuring 8 operational views, priority purchase order worklists, what-if simulation, and an AI intelligence assistant.

2. Business Problem & Objectives

Core Business Pain Points:

- **Lost Revenue from Stockouts:** High-velocity staples frequently experienced zero stock, causing immediate revenue loss and unrecorded customer demand.
- **Trapped Working Capital:** Stagnant inventory tied up over Rs. 1.78B in capital exceeding 120+ days of supply.
- **Flawed Historical Heuristics:** Standard rolling averages under-forecasted items recovering from stockouts.
- **Lack of Actionable Priority:** Supply chain planners lacked automated, priority-ranked purchase order and stock transfer workflows.

3. Enterprise Dataset Architecture

Dataset	Records	Grain / Key	Attributes & Role
transactions_clean	9,945,396	Transaction x Date	Sales quantities, prices, discounts, promos, channels (2022-2025)
inventory_clean	26,408	Store ID x SKU ID	Stock on hand, safety stock, reorder point, last restock date
products_clean	5,000	SKU ID	SKU name, 12 categories, brand, cost price, retail price
stores_clean	30	Store ID	Store name, 4 store types, 4 metropolitan cities, square footage
sku_inventory_flags	11,649	Store ID x SKU ID	Synthetic stockout and slow mover flags (Validation benchmark only)

4. Forecasting Model Benchmarks & LightGBM Results

All models were evaluated on the strict out-of-sample chronological test set (H2 2025: 1,066,548 daily observations) using 52 engineered rolling, lag, and calendar features.

Model Architecture	Test WAPE	Test RMSE	Test MAE	Test Bias	Error Reduction
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Historical Global Mean	68.42%	2.140	1.482	+4.12%	Baseline
Lag-1 Persistence	58.19%	1.812	1.261	-0.15%	-14.9%
Lag-7 Seasonal Lag	52.34%	1.624	1.134	-0.08%	-23.5%
28-Day Rolling SMA (Best Baseline)	47.54%	1.485	1.030	-0.04%	-30.5%
LightGBM Model 1 (L1 Loss)	43.82%	1.246	0.949	-0.84%	-35.9%
LightGBM Model 2 (Production L2)	42.16%	1.183	0.928	-0.01%	-38.4% (+11.3% vs SMA)

5. Inventory Risk Scoring & Working Capital Audit

Applying the production forecasts across 26,408 positions revealed extreme inventory polarization across NorthBay Living's network:

Risk Classification Tier	Position s	% Network	Stock Units	Trapped Capital (Cost Basis)	Operational Recommendation
CRITICAL_STOCKOUT	4,239	16.05%	0	Rs. 0.00	Emergency expedited purchase order
HIGH_STOCKOUT_RISK	970	3.67%	48,192	Rs. 19,420,110	Preemptive PO to restore safety buffer
MEDIUM_STOCKOUT_RISK	764	2.89%	61,208	Rs. 24,810,400	Standard lead-time replenishment PO
HEALTHY_OPTIMAL	2,444	9.25%	137,497	Rs. 70,781,417	Maintain routine monitoring
MEDIUM_OVERSTOCK	1,899	7.19%	298,401	Rs. 122,110,950	Reduce next cycle order quantity
HIGH_OVERSTOCK	6,708	25.40%	1,421,801	Rs. 598,340,120	Freeze purchase orders & reallocate stock
CRITICAL_OVERSTOCK	9,384	35.53%	2,421,800	Rs. 1,057,994,569	Execute clearance markdowns (25%-35% off)
TOTAL NETWORK	26,408	100.0%	4,388,899	Rs. 1,849,231,166	Enterprise Working Capital Optimization

6. Operational Decision Platform & Delivery

The project delivered an 8-view interactive intelligence dashboard (app.py) incorporating:

- **Executive Overview:** Macro-level KPI scorecards, working capital distribution, and department exposure.
- **Priority Action Center:** Automated, exportable worklists for emergency POs, safety buffer restoration, clearance sales, and cross-store transfers.
- **What-If Simulator:** Real-time scenario planner allowing planners to test variable supplier lead times, stock buffers, and demand shifts.
- **Intelligence Assistant:** Natural-language data exploration engine with single-SKU explainability.

7. Key Business Findings & Recommendations

1. **Rs. 1.78 Billion Working Capital Liquidation:** 68.1% of inventory positions exceed healthy thresholds, locking up Rs. 1.78B. Immediate clearance campaigns can free up hundreds of millions of rupees in liquid capital.
2. **Preventing 5,209 Out-of-Stock Deficits:** Proactive replenishment using Model 2 demand forecasts eliminates lost sales on 19.7% of products.
3. **Cross-Store Rebalancing:** Leveraging automated inter-store stock transfers reduces overall procurement expenditures by moving excess stock from overstocked stores to out-of-stock locations.

8. Project Verification Links

- **GitHub Repository:** <https://github.com/MHK-123/Foresight>
- **Web Application Source:** `app.py`
- **Model Artifact:** `models/lightgbm_l2_demand_forecast.joblib`
- **Scored Inventory Dataset:** `risk_engine_outputs/inventory_risk_scored.parquet`
- **Governance Documents:** `PRIVACY_POLICY.md` | `TERMS_OF_SERVICE.md`